

EXP 06

Shared Memory IPC

AIM:

To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

ALGORITHM

1. Start
2. Include required header files
3. Create a shared memory segment using shmget()
4. Attach shared memory using shmat()
5. Get message from sender process
6. Store message in shared memory
7. Receiver process attaches same shared memory
8. Receiver reads message from shared memory
9. Detach shared memory using shmdt()
10. Remove shared memory using shmctl()
11. End

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/ipc.h>
#include <sys/shm.h>
```

```
int main() {
    int shmid;
    key_t key = 1234;
    char *shared_memory;
    char message[100];
```

```
    shmid = shmget(key, 1024, 0666 | IPC_CREAT);
```

```
    if (shmid == -1) {
        printf("Shared memory creation failed\n");
```

```

    exit(0);
}

shared_memory = (char *)shmat(shmid, NULL, 0);

if (shared_memory == (char *)-1) {
    printf("Shared memory attachment failed\n");
    exit(0);
}

printf("Enter message to write into shared memory: ");
fgets(message, sizeof(message), stdin);

strcpy(shared_memory, message);

printf("\nData written into shared memory successfully.\n");
printf("Data read from shared memory: %s", shared_memory);

shmdt(shared_memory);

shmctl(shmid, IPC_RMID, NULL);

return 0;
}

```

RESULT:

Thus, Inter Process Communication using shared memory was successfully implemented and executed.